

Dipayan Biswas

Applied Scientist • Computer Vision & Multimodal AI (VLMs)

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Summary

PhD researcher in Computer Science with 5+ years in multimodal AI and computer vision; published at IEEE venues. Builds LLM/VLM-powered video understanding and summarization systems, with cross-modal fusion and LLM-as-judge evaluation.

Education

University of Houston (UH)

Houston, TX

Ph.D. in Computer Science — Expected Dec 2026

Khulna University of Engineering & Technology (KUET)

Khulna, Bangladesh

B.Sc. in Electronics and Communication Engineering — University Gold Medalist (2012–2016)

Technical Skills

Specializations: Computer Vision, VLMs, Multimodal AI, Cross-modal Fusion, Multimodal Retrieval, LLM Evaluation

MLOps: FastAPI, vLLM, Ollama, Docker, RAG, MLflow, DeepEval, Hydra, AWS (SageMaker, EC2, S3), Git

Frameworks: PyTorch, HuggingFace Transformers, TensorFlow, OpenCV, LangChain, Scikit-learn

Languages: Python, C++, SQL, JavaScript

Experience

University of Houston — Graduate Research Assistant

Aug 2021 – Present

- Led multimodal research on the NSF-funded VideoPoints platform, distilling hour-long lectures into hierarchically structured summaries with supporting visuals, deployed across 10+ university courses serving 750+ students.
- Designed cross-modal fusion over ASR transcripts, OCR text, and video frames via VLMs and in-context learning, driving both summarization and selection of representative visuals spanning key concepts.
- Built automated LLM-as-judge evaluation benchmarks (DeepEval, OpenAI/Gemini APIs), raising factual faithfulness to 96% (+9.6) and relevance to 84% (+5.2) over text-only baselines.
- Improved object detection from 90.8% to 95.3% AP50 via YOLO11 + semi-supervised auto-labeling (IEEE MIPR 2025).
- Proposed a novel graph-based method grouping disjointed entities into unified objects (+15.8% mAP, IEEE ISM 2023).

Fellowship.AI — Machine Learning Fellow

Sep 2025 – Nov 2025

- Developed a multimodal recommendation system (Gemini API) that jointly analyzes listing images and descriptions to deliver structured insights for pricing, staging, and visual content optimization.
- Built a pipeline over 8K+ Redfin listings with weighted KNN and outlier detection to flag mispriced properties.

KUET — Lecturer / Assistant Professor

Feb 2017 – Aug 2021

- Developed deep-learning and ensemble models for plant leaf disease, chest X-ray pneumonia, and breast cancer detection.
- Mentored 10+ undergraduates on ML research; resulting work published at IEEE EICT and ICAEE.

Samsung R&D Institute Bangladesh (SRBD) — Software Engineer

Jul 2016 – Feb 2017

- Built panoramic, 360-degree, and little-planet rendering modes for visualizing captured photos and videos in the Samsung Gear 360 iOS app (OpenCV, OpenGL), improving image quality and reducing latency.

Selected Projects

Unified VLM Inference Framework [\[Code\]](#)

Mar 2026 – Apr 2026

- Engineered unified VLM inference across 5+ backends (Ollama, vLLM, HuggingFace, OpenAI/Gemini APIs).
- Designed a shared interface for interleaved text-image prompting and multi-step workflows, portable across backends.

Lecture Video Chaptering

Jun 2025 – Aug 2025

- Devised a chaptering pipeline aligning Whisper transcripts with slide transitions to segment long lecture videos.
- Applied boundary validation and correction to enforce contiguous, gap-free chapter boundaries.

Small & Distant Face Detection [\[Code\]](#)

Feb 2023 – May 2023

- Extended RetinaFace with an auxiliary IoU head and multi-task loss to detect small, distant, and occluded faces.
- Trained ResNet50 and MobileNet0.25 backbones on WIDER FACE, improving small/distant-face robustness.

Honors, Certifications & Leadership

- Honors: Ph.D. Research Showcase Runner-Up, UH • Cullen Graduate Fellowship • University Gold Medalist, KUET.
- Certifications: Generative AI with LLMs • Machine Learning in Production • Building & Evaluating Advanced RAG
- Leadership: Organizing Secretary, Graduate Student Association of Bangladesh, UH (2022–2025).